

SATO Report

Page Budget

DAC Rule: up to 6 pages plus an additional 1 page for references only

Page 1 : Abstract + Intro

Problem, gap, solution, 3 contribution bullets

Page 2: Background + Motivation

Neuron model, how existing accelerators work, 3 numbered limitations

Page 2-5: Their Architecture details

Dataflow, bucket sort dispatcher, hardware details

Page 5-6: Experiments

Setup, 3 result sections, 2 sensitivity studies

Page 6: Conclusion and references

One paragraph conclusion, 21 references

How they present their story

Fig 1: basic spiking neuron

Fig 2: conventional dataflow vs their dataflow

Fig 3,4,5: their system arch, PE arch, their workload dispatch strategy

Figs 7,8,9: energy and performance, PE & timestep scaling

Experiments and metrics

Tech node: 28 nm, 200 MHz

Hardware flow: Verilog PE, Synopsys Design Compiler, CACTI

Algorithm validation: PyTorch

Energy breakdown and latency: normalized energy across all 5 networks * 4 accelerators, split into global Buffer, DRAM, core and total execution cycles

Power and area: component level breakdown, reporting pwr and area split for each

Accelerator comparison:

Eyeriss, Eyeriss-SNN, S2N2(an fpga snn acc), SpinalFlow

Tech node, frequency, PEs, area, power, throughput in GOP/s, area efficiency GOP/s/mm²

PE Scaling: 128 to 1024 PEs on TSC-SNN only, speedup and energy vs SpinalFlow

result paragraphs follow a fixed pattern, the number, then the reason. "SATO consumes 91.3%, 83.4% and 69.7% less energy than Eyeriss, S2N2 and SpinalFlow. This is mainly due to..."